

Question	Answer	Mark	Mark scheme	Additional guidance
12	$\frac{4}{9}$	P1 P1 A1	for process to find link between volume of Q and volume of P or between volume of R and volume of Q, eg ratio 1.5 : 1 or $Q = 1.5P$ or $P = \frac{2}{3}Q$ or two values in the ratio 1 : 1.5 such as 100 and 150 for process to find link between volume of R and volume of P eg $1.5^2 : 1$ or two values in the ratio 1 : 2.25 such as 100 and 225 for $\frac{4}{9}$ oe fraction eg $\frac{100}{225}$	$1.5^2 (= \frac{9}{4})$ is enough for this mark, award P1P1 Accept $P = \frac{4}{9}R$